Vascular Development and Differentiation During Human Liver Organogenesis

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ABSTRACT

The vascular architecture of the human liver is established at the end of a complex embryological history. The hepatic primordium emerges at the 4th week and is in contact with two major venous systems of the fetal circulation: the vitelline veins and the umbilical veins. The fetal architecture of the afferent venous circulation of the liver is acquired between the 4th and the 6th week. At the end of this process, the portal vein is formed from several distinct segments of the vitelline veins; the portal sinus, deriving from the subhepatic intervitelline anastomosis, connects the umbilical vein, which is the predominant vessel of the fetal liver, to the portal system; the *ductus venosus* connects the portal sinus to the vena cava inferior. At birth, the umbilical vein and the *ductus venosus* collapse; the portal vein becomes the only afferent vein of the liver. The efferent venous vessels of the liver derive from the vitelline veins and are formed between the 4th and the 6th week. The hepatic artery forms at the 8th week; intrahepatic arterial branches progressively extend from the central to the peripheral areas of the liver between the 10th and the 15th week. Hepatic sinusoids appear very early, as soon as hepatic cords invade the septum transversum at the 4th week. They then progressively acquire their distinctive structural and functional characters, through a multistage process. Vascular development and differentiation during liver organogenesis is, therefore, a unique process; many of the cellular and molecular mechanisms involved remain poorly understood. Anat Rec, 291:614–627, 2008. © 2008 Wiley-Liss, Inc.

Key words: liver; fetal liver; vitelline veins; umbilical veins; portal vein; portal sinus; *ductus venosus*; terminal hepatic veins; *ligamentum teres*; *ligamentum venosum*; hepatic artery; peribiliary plexus; hepatic sinusoids

The vascular architecture of the human adult liver is particularly complex. Two distinct afferent circulatory systems coexist: one, represented by the portal vein and its intrahepatic branches, is responsible for the functional circulation of the liver, which conveys a blood of venous origin, coming from the intestine, the pancreas and the spleen; the other, represented by the hepatic artery and its intrahepatic branches, is responsible for the nutritional circulation of the liver and transports a blood of arterial origin; the portal system is the predominant

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TABLE 1. Comparative chronology of the developmental stages of the liver in human, rat and mouse embryos

Carnegie stage	Characteristics	Days		
		Human *mean (range)*	Rat	Mouse
11	Formation of the hepatic primordium	24 (23-26)	11.5	10
12	Development of the hepatic diverticulum Appearance of the hepatic trabeculae in the septum transversum	28 (26-30)	12	10.5
13	Development of the hepatic diverticulum Proliferation of the hepatic cords emmeshing the sinusoids Collapse of the right umbilical vein	30 (28-32)	12.5	11
14	Rapid growth of the liver enlage "Symmetrical stage" of the fetal circulation of the human liver	33 (31-35)	13	11.5
15	Reorganization of the fetal circulation of the human liver	36 (35-38)	13.5	12
16	Establishment of the definitive "fetal" circulation of the human liver	40 (37-42)	14	12.5
17		42 (41-44)	14.5	13
18	Formation of the ductal plates	45 (44-48)	15	13.5
19		49 (48-51)	15.5	14
20		52 (51-53)	16	14.5
21		54 (53-55)	16.5	15
22		55 (55-56)	17	15.5
23	Formation of the intrahepatic biliary ducts	57 (56-58)	17.5	16

one and, in humans, is responsible for approximately 75% of the total liver blood inflow in the normal state (Zwiebel et al., 1995). Only one efferent circulatory system is present: it is formed by intrahepatic efferent veins of increasing grades, which eventually drain, through one of the three suprahepatic or terminal veins, into the inferior vena cava. Between the afferent and efferent vessels, several independent intrahepatic capillary systems coexist, in close association with the different microanatomical compartments of the liver: several microvascular networks are found in the portal spaces, such as the peribiliary plexus and the periportal plexus, whereas the capillary system associated with the hepatic lobules is formed by highly specialized microvessels, known as liver sinusoids, which control the bidirectional exchanges between hepatocytes and the blood and drive the functional heterogeneity of liver parenchyma. The various vascular compartments of the liver are lined by different subpopulations of endothelial cells, which present distinctive structural and functional characteristics, culminating with the highly differentiated phenotype displayed by the endothelial cells lining hepatic sinusoids.

The fetal circulation of the liver is very different from the adult one. From a structural point of view, it is even more complex. Indeed, both the macro- and the microcirculations of the fetal liver look like a puzzle formed by elements of distinct origins which remain at long independent and only eventually establish definitive connections to give rise to the adult vascular pattern. The fetal liver receives two different afferent venous vessels, the portal vein and the umbilical vein, which derive from two distinct extraembryonic systems, the umbilical veins and the vitelline veins, and have to undergo major changes during the embryonic life and then at birth before the definitive venous system of the adult liver is established. The evolution of the intrahepatic arterial system is, in part, independent and disconnected from that of the intrahepatic veins. Finally, the main capillary networks of the adult liver have independent origins, because liver sinusoids derive from a totally different vascular system than the intraportal capillary networks.

The differences between the fetal and the adult circulations of the liver are not only structural, but also functional. The fetal liver exerts specific functions, such as hematopoiesis, which are not maintained in the adult liver; in parallel, the fetal liver progressively acquires the full range of physiological and metabolic competences displayed by the postnatal and adult liver. In consequence, during the fetal life, intrahepatic vessels face two different constraints: (1) they must fulfil all the structural and functional requirements necessary to contribute to the specific roles exerted by the fetal liver; (2) they must acquire the structural and functional characteristics which will be necessary to respond to the functions exerted by the postnatal liver. These two parallel processes explain the successive, and often complex, stages of structural and functional differentiation displayed by fetal vessels and their lining populations, especially endothelial cells.

Original and first-hand data regarding the stages of development of the vascular architecture of the liver are relatively scarce. For a long period of time, they were based only on microanatomical studies of embryos and fetuses obtained by chance: the material was rare, its analysis was difficult and a substantial amount of interpretation was required (Barclay et al., 1944). Renewed interest for the study of fetal hepatic vessels came from the development of ultrasound investigations which have made it possible to test the validity of previous anatomical concepts, to better evaluate the degree of interindividual variability and to provide fresh hemodynamic data to substantiate the previous morphological findings (Morin and Winsberg, 1978; Champetier et al., 1989). Moreover, animal models have provided new data on the mechanisms of vascular development and differentiation during liver organogenesis: at least some of them are transferable to the human situation (for a correspondence between the developmental stages of the liver in human, rat and mouse embryos, see Table 1).

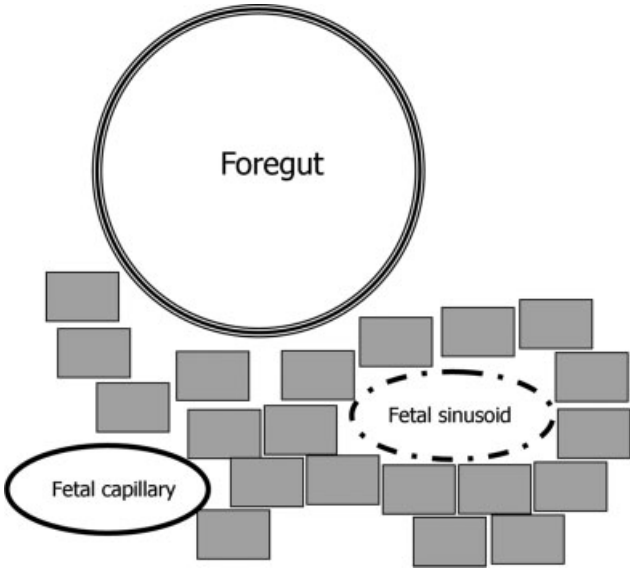


Fig. 1. Schematic representation of the hepatic diverticulum, with cords of primitive hepatocytes budding from the foregut, invading the mesenchyme of the septum transversum, and emmeshing the pre-existing fetal vessels.

HEPATIC PRIMORDIUM AND ITS INTERACTIONS WITH EMBRYONAL VESSELS DURING THE EARLY STEPS OF LIVER ORGANOGENESIS

The first evidence of liver development can be traced during the 4th gestational week (Carnegie embryonal stage 11, Table 1), as a local thickening of the endoderm lining the ventral wall of the very distal part of the foregut, where the fetal intestine meets the yolk sac (Hutchins and Moore, 1988; Godlewski et al., 1997). At the end of the 4th gestational week (Carnegie embryonal stages 12 and 13, Table 1), the proliferative activity of the cells of the hepatic primordium results in the development to the hepatic diverticulum, which buds from the ventral wall of the foregut (Fig. 1). The extremity or cranial part of the hepatic diverticulum is formed by a small mass of epithelial cells: it will give origin to the liver. The extrahepatic bile ducts, the cystic duct and the gallbladder will develop from the caudal part of the hepatic diverticulum, which, at this stage, presents as a hollow tube.

The epithelial cells of endoblastic origin forming the early liver bud rapidly invade the caudal part of the *septum transversum*, an area of mesenchymal tissue located in the precardiac region, in which they form anastomosed cords separated by vascular channels. The mesenchyme of the *septum transversum* will give origin to the connective tissue associated with the intrahepatic vessels, including sinusoids and portal vessels, and to the connective capsule of the liver, the Glisson's capsule. The liver bud is rapidly growing: at the end of the 6th gestational week (Carnegie embryonal stage 16, Table 1), the fetal liver occupies most of the abdominal cavity (Hutchins and Moore, 1988) (Fig. 2).

Because of its anatomical situation, the fetal liver is in close contact with most of the major fetal vessels (Fig. 3). At the end of the 3rd week of gestation (Carnegie embryonal stage 10), three major venous systems are present in the embryo: two are extraembryonic, the vitelline veins and the umbilical veins, one is intraembryonic, the cardinal veins (Kiserud and Acharya, 2004). The paired vitelline veins, right and left, transport blood from the yolk sac to the heart. The initially paired umbilical veins, right and left, transport oxygenated blood from the placenta to the heart, through the hepatic primordium. The cardinal veins drain the whole venous blood of the embryo to the heart. All these three major venous systems converge into the *sinus venosus*, a large quadrangular cavity, which corresponds to the caudal part of the fetal heart and which will be later incorporated in the cardiac wall (Kiserud and Acharya, 2004). Major changes will then occur in the distribution and arrangement of the major venous systems associated with the liver, leading to the definitive fetal vascular pattern and then, to the definitive adult pattern.

AFFERENT VENOUS VESSELS OF THE FETAL LIVER

In contrast to the adult liver, which receives only one large afferent venous vessel, the fetal liver receives two different afferent venous vessels, originating from extraembryonic circulations: one results from the evolution of the left umbilical vein and the other derive from the vitelline veins. A first sequence of changes occurs between the 4th and the 6th gestational week (Carnegie embryonal stages 11 to 16, Table 1), to achieve the "definitive" fetal pattern of hepatic vascular architecture. A second major sequence of changes occurs at birth and gives origin to the definitive adult pattern of hepatic vascular architecture.

"Symmetrical Stage": Extraembryonic Venous Vessels and the Liver

During the 4th gestational week, very soon after the emergence of the liver bud, both the vitelline and the umbilical systems undergo major changes (Dickson, 1957; Fig. 4).

Development of Intervitelline Anastomoses

The two vitelline veins become connected to one another by several anastomoses (Dickson, 1957; Joyce and Howard, 1988; Howard, 1999). In the prehepatic segment, three anastomoses are classically described: (1) the most caudal one is the inferior intervitelline anastomosis, located on the ventral side of the distal segment of the duodenum, formed in the 5-mm embryo; (2) the middle intervitelline anastomosis, also called the retroduodenal or dorsal anastomosis, is the first one to appear, in the 3.5- to 4-mm embryo, and is located on the dorsal side of the duodenum; (3) the superior or subhepatic anastomosis is located under the hepatic primordium, on the ventral side of the duodenal segment. A fourth intervitelline anastomosis, the subdiaphragmatic anastomosis, is located above the hepatic primordium and immediately under the *sinus venosus*.

The segments of the right and left vitelline veins located between the subdiaphragmatic and the subhepatic anastomoses become interrupted by the developing

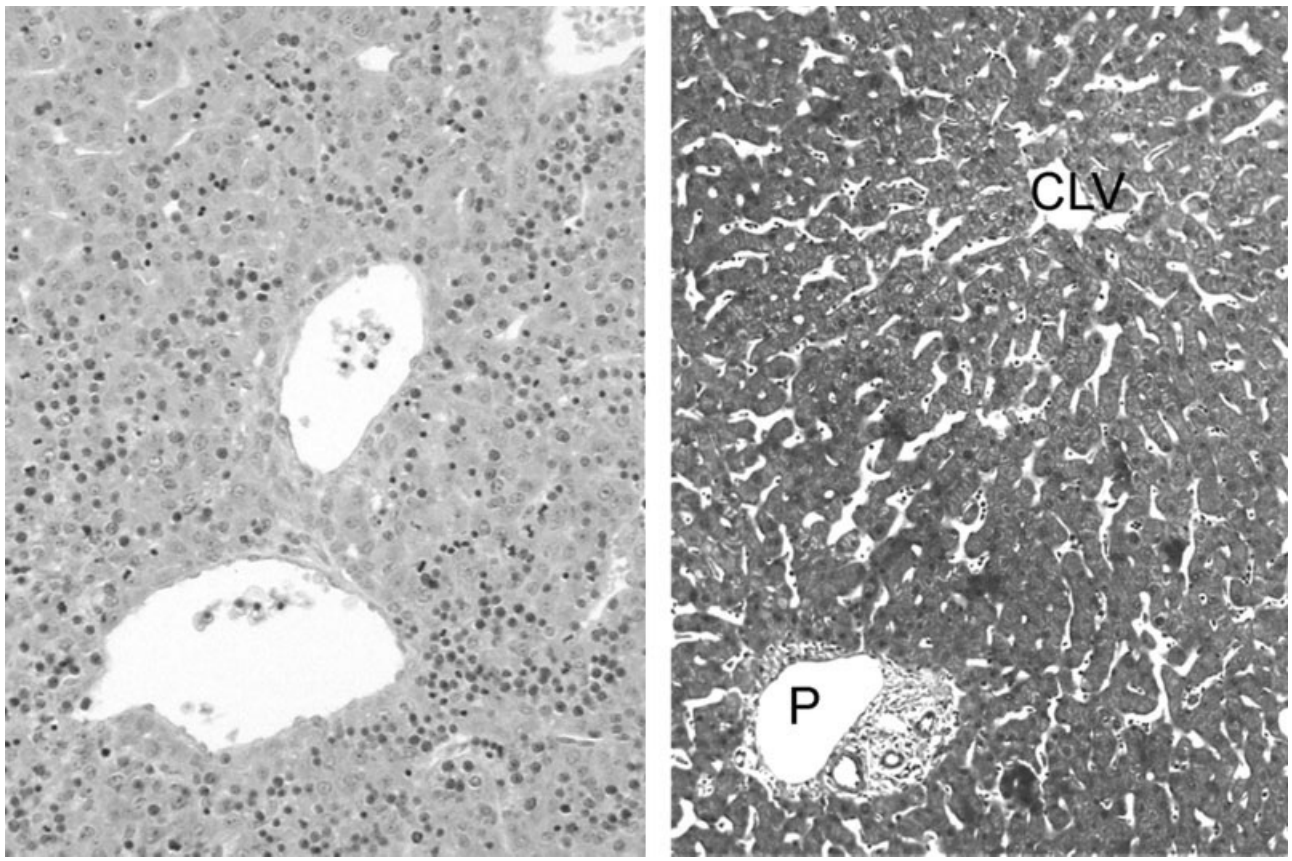


Fig. 2. Histological view of the embryonic (left panel) and adult (right panel) human liver. Embryonic liver: large intrahepatic veins are running within the hepatic parenchyma, formed by large plates of hepatocytes separated by sinusoids containing numerous hematopoietic cells (HES, original magnification: ×180). Adult liver: low power view of a hepatic lobule, limited by a portal space (P) and centered by a centrilobular vein (CLV); hepatocyte plates are separated by well formed sinusoids (HES, original magnification: ×120).

hepatic trabeculae and resolve into a plexus of small vessels connected with the sinusoids. Only the most caudal and the most cranial segments of this portion of the vitelline veins will persist. Small venous branches, named *venæ advehentes*, convey the blood from the subhepatic anastomosis to the sinusoidal plexus; in the same way, small venous vessels, or *venæ revehentes*, drain the blood of the sinusoidal plexus into the subdiaphragmatic anastomosis, from which it passes to the *sinus venosus*.

Transfer of the Umbilical Circulation to the Liver

Meanwhile, the umbilical vein system also undergoes a major remodeling. Until this stage, both the right and left umbilical veins are formed by a prehepatic trunk, which divides into two branches, one ending directly into the liver parenchyma and the other one running along the liver to open into the corresponding angle of the *sinus venosus*. The terminal branches of the right umbilical vein collapse at the 4th week of gestation (Carnegie embryonal stage 13, Table 1); the prehepatic segment of the vein persists but directly ends into the left umbilical vein within the abdominal wall (Dickson, 1957; Lassau and Bastian, 1983). Meanwhile, the left umbilical vein undergoes major changes. Its terminal branch directed to the *sinus venosus* collapses. Only the hepatic branch persists. It greatly enlarges, continues to grow within the liver parenchyma and eventually makes a connection with the left angle of the subhepatic intervitelline anastomosis. At this stage, the umbilical circulation is, therefore, completely transferred to the liver. This finding suggests that all the oxygenated blood conveyed by the remaining umbilical vein has to pass through the liver before reaching the heart from which it will be redistributed to the whole embryo.

From the "Symmetrical" to the "Asymmetrical" Stage: The Establishment of the Afferent Venous Circulation of the Embryonic Liver Between the 4th and the 6th Gestational Week

This period, beginning at the 5th week of gestation and ending at the 6th week, is marked by the emergence of the portal vein from the vitelline system, the remodeling of the umbilical system through the development of the *ductus venosus* and the establishment of a permanent connection between the portal and the umbilical circulations through the *sinus intermedius* or portal sinus (Fig. 5). At the end of this complex process, only two large, unpaired and interconnected afferent venous

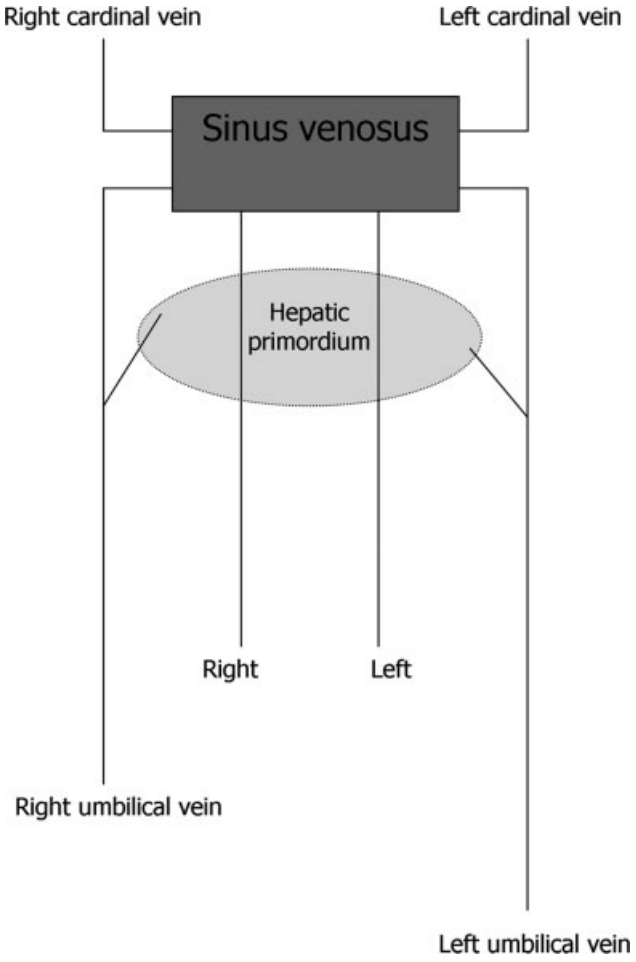


Fig. 3. Schematic representation of the relations between the hepatic primordium and the three major venous systems of the fetal circulation at the beginning of the 4th week of gestation: the vitelline veins, the umbilical veins and the cardinal veins; all three systems converge into the *sinus venosus*.

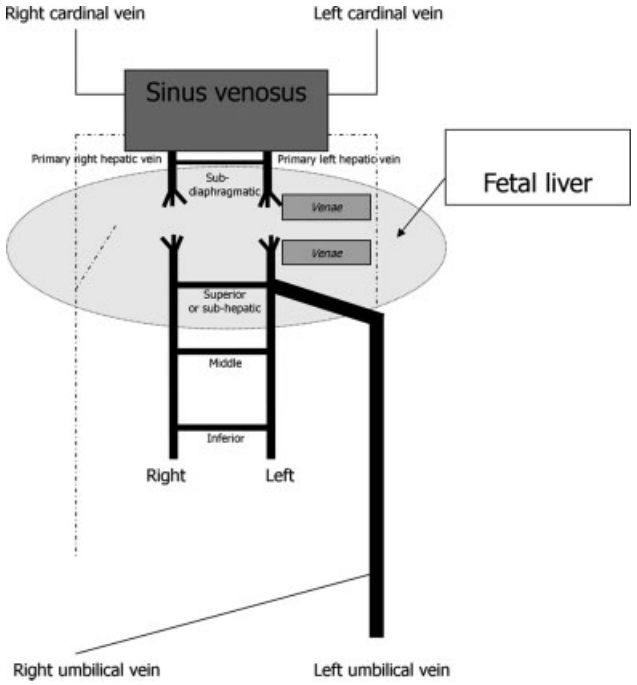


Fig. 4. The so-called "symmetrical stage" of the fetal circulation of the liver at the end of the 4th week of gestation. The two vitelline veins are connected through four distinct intervitelline anastomoses (IVA). The growing liver has disrupted the segments of vitelline veins located between the subhepatic and the subdiaphragmatic anastomoses. The afferent vessels are located on the caudal side and are connected through the *venæ advehentes* with the sinusoidal plexus. The efferent vessels are located on the cranial side and are fed by the *venæ advehentes*, draining the sinusoidal plexus. The terminal branches of the right umbilical vein have collapsed. The left umbilical vein terminates into the left angle of the subhepatic intervitelline anastomosis, from which two large intrahepatic vessels emerge, including the *ramus angularis* on the left side.

vessels will penetrate the embryonic liver: this is the so-called asymmetrical stage, which will persist until birth.

Development of the Portal Vein

The mature portal vein is usually assumed to derive from various segments of the previous right and left vitelline veins: this is the so-called vestigial theory (Dickson, 1957; Marks, 1969; Joyce and Howard, 1988). From its caudal to its cranial extremities, the trunk of the portal vein is formed by: (1) a segment of the left vitelline vein, from the termination of the mesenteric vein to the middle intervitelline anastomosis; (2) the middle or retro-duodenal intervitelline anastomosis; (3) the segment of the right vitelline vein comprised between the retro-duodenal and the subhepatic anastomoses. The other segments of the right and left vitelline veins collapse and disappear. The complex origin of the mature portal vein explains its typical S-shape along the duodenum. The so-called vestigial theory of the formation of the portal vein has been criticized by some authors, mainly on the basis that vitelline veins are essentially non functional in the human embryo, which has a chorioallantoic placenta, in contrast to the embryos of many other mammals, which have a chorio-vitelline placenta (Lassau and Bastian, 1983).

The portal vein ends into the right angle of the subhepatic anastomosis. This anastomosis will soon transform into a large intrahepatic vascular channel, known as the *sinus intermedius* or portal sinus (Mavrides et al., 2001). From the upper (or cranial) side of the right and left angles of the subhepatic anastomosis, two large intrahepatic venous vessels emerge. The left vessel is known as the *ramus angularis* (Dickson, 1957). Several large branches emerge from the trunk of the portal vein and from the tributaries of the portal sinus and give origin to vascular channels of decreasing diameters; the smallest ones are connected with the sinusoidal network. This is the basis for the future intrahepatic portal circulation.

All the intrahepatic segments of the portal vein and its branches become surrounded by a rim of mesenchymal tissue, from which the portal spaces will form and within which the intrahepatic branches of the hepatic artery will later grow. However, it must be underlined that, at this stage, the portal venous system is not the predominant afferent system of the liver: most of the

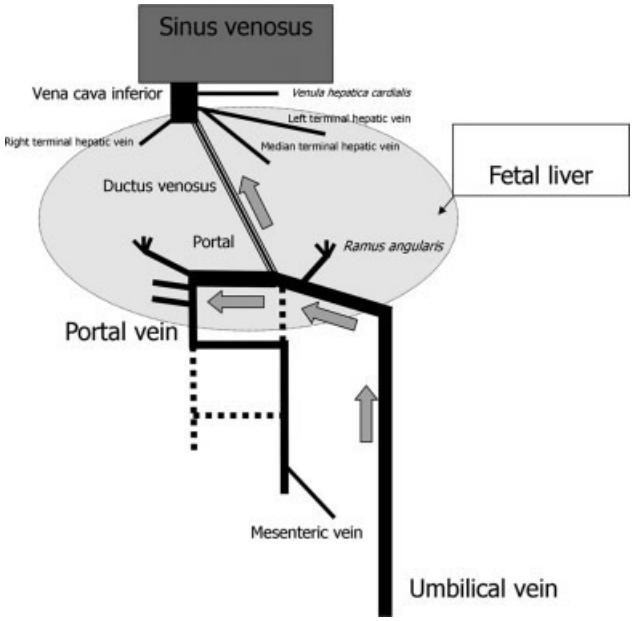


Fig. 5. The "definitive" fetal circulation of the liver, at the 6th week of gestation. The portal vein is formed by several segments of the previous vitelline veins. The subhepatic anastomosis has given origin to the portal sinus, which connects the umbilical vein to the portal system. The *ductus venosus* connects the left angle of the portal sinus to the vena cava inferior, which also receives the termination of the three efferent terminal hepatic veins. The arrows indicate the direction of the predominant blood flow, coming from the umbilical vein; the largest part of the blood flows into the portal sinus to fed the sinusoidal plexus; the remaining is shunted to the vena cava inferior by the branchless *ductus venosus*.

liver blood inflow comes from the umbilical vein, which, meanwhile, has also undergone important changes.

Remodeling of the Umbilical Circulation, the Formation of the Portal Sinus and the Development of the *ductus venosus*

During the fetal life, the umbilical vein is the predominant afferent vessel of the liver. It derives from the original left umbilical vein and runs into the abdominal cavity, from the umbilicus to the embryonic liver, along the falciform ligament. After the transfer of the umbilical circulation to the liver, the main trunk of the umbilical vein ends directly into the left angle of the subhepatic intervitelline anastomosis, near the emergence of the *ramus angularis*.

Between the 4th and the 6th week of gestation, the subhepatic intervitelline anastomosis transforms into a large, definitive intrahepatic channel connecting the intrahepatic portal vein and the umbilical vein, and known as the *sinus intermedius* or portal sinus (Mavrides et al., 2001; Paris et al., 2004). This transformation results from the regression of the left half of the subhepatic anastomosis, which is likely incorporated into the umbilical vein, as suggested by the apparent displacement of the *ramus angularis* from the left angle of the subhepatic anastomosis to a point located caudally along the umbilical vein (Dickson, 1957). The remaining portion of the subhepatic anastomosis enlarges to form the portal sinus, which connects the umbilical circulation with the portal venous system (Mavrides et al., 2001). This large connection makes it possible for a substantial part of the oxygenated blood conveyed by the umbilical vein to be delivered directly to the sinusoidal network, and then, to hepatocytes. It has been claimed that the blood flow conveyed by the umbilical vein to the hepatic parenchyma plays a major organizational role, contributing, through hemodynamic constraints, to the regular orientation of the sinusoids and to the progressive laminar distribution of the hepatic trabeculae located between the sinusoids; this process likely begins in the left liver and progressively involves the right liver (Lassau and Bastian, 1983).

The left angle of the portal sinus is connected to the inferior vena cava by a specialized vessel, known as the *ductus venosus*, which is in the prolongation of the umbilical vein, but not in direct connection with it (Mavrides et al., 2001, 2002). The *ductus venosus* is a large, branchless vascular channel running through the liver parenchyma without giving origin to any intrahepatic branch. Its wall is rich in elastic fibers and contains smooth muscle cells (Mavrides et al., 2002); the existence of a muscular sphincter at the origin of the *ductus venosus* is controversial (Gennser, 1992), but concurrent histological studies suggest the existence of muscular pillows at the ostium of the vessel (Mavrides et al., 2002; Ailamazyan et al., 2003). In vivo hemodynamic studies confirm that a certain regulation of blood flow through this vascular channel can take place. The *ductus venosus* plays a very important physiological role: it behaves as a shunt allowing for a substantial part of the total umbilical blood flow to bypass the liver to be delivered directly to the heart from which it is redistributed to the whole embryo (Tchirikov et al., 2006). This part has been evaluated to 50% in most animals, but seems to be lower in humans: 30% at 20 weeks of gestation and approximately 20% at 30 weeks (Kiserud et al., 2000; Pennati et al., 2003).

The early development of the *ductus venosus* is difficult to trace and to reconstitute. Several hypotheses have been proposed (Dickson, 1957): (1) according to early works, this vessel results from the coalescence and enlargement of a group of sinusoids located between the subhepatic and subdiaphragmatic anastomoses: this concept is now largely abandoned; (2) according to more recent works, the *ductus venosus* develops from a small vessel connecting the subhepatic and subdiaphragmatic anastomoses; this vessel originates from the middle part of the subhepatic intervitelline anastomosis; it rapidly enlarges after the disappearance of the intrahepatic left and right vitelline veins and the transfer of the whole umbilical circulation to the liver; it position shifts to the left angle of the emerging portal sinus after the regression of the left half of the primitive subhepatic anastomosis (Dickson, 1957). In the same way, the terminal portion of the developing *ductus venosus* follows the complex history of the formation of the inferior vena cava. This large intraembryonic vessel results from the merging of the right angle of the sinus venosus and of the corresponding part of the subdiaphragmatic anastomosis between the primitive vitelline veins.

By the end of the 6th gestational week (Carnegie embryonal stage 16, Table 1), the intrauterine pattern of the hepatic circulation is established. No major change

will occur, except for the progressive displacement of the *ramus angularis* toward the origin of the intrahepatic umbilical vein, which will be reached at the end of the embryonic period (Dickson, 1957). All the major vascular structures (portal vein, umbilical vein, portal sinus, *ductus venosus*) can be easily discerned and studied at ultrasound examinations during gestation (Mavrides et al., 2001), and their structural and functional abnormalities can be diagnosed (Gallego et al., 2002, 2004). Hemodynamic studies performed during the intrauterine life have confirmed that, until birth, the umbilical vein is the predominant liver vessel, accounting for approximately 80% of the liver blood inflow; there is an asymmetrical distribution of blood flow between the left and the right liver lobes; almost 100% of the blood inflow of the left lobe comes from the umbilical vein, in contrast to 50% of that of the right lobe, the remaining 50% coming from the portal vein (Haugen et al., 2004).

Acquisition of the Definitive Vascular Pattern of the Liver: Changes in the Afferent Venous Vessels of the Liver at Birth

Major changes occur at birth in the umbilical circulation and, indirectly, in the intrahepatic portal system (Fig. 6). At the end of this process, the umbilical circulation will collapse whereas the portal vein will remain the only afferent venous system of the liver: the adult pattern of the hepatic vascular architecture will be established.

The process of closure of the umbilical circulation begins immediately after birth. The two umbilical arteries collapse first. The umbilical vein usually closes a little after the umbilical arteries: its blood flow is interrupted and its blood content begins to clot. The process begins in the first minutes after birth, but is not complete until 15 to 20 days. The brutal decrease in blood flow and pressure in the umbilical vein results in the complete collapse of the *ductus venosus*; its permanent closure is due to the obliteration of the vascular channel by deposition of a connective tissue; the process starts within days after birth and is completed between 1 and 3 months of age (Meyer and Lind, 1966; Edelstone, 1980).

In the same time, the collapse of the umbilical circulation induces a reversal of the direction of the blood flow within the portal sinus: instead of flowing predominantly from the umbilical vein to the portal system, the blood now flows freely from the portal vein into the portal sinus and its remaining permeable tributaries [according to some authors (Lassau and Bastian, 1983), it is even only at this stage that the portal vein becomes really functional, but this is in contrast with the results of more recent in vivo hemodynamic studies]. In any case, the collapse of the umbilical circulation results in the definitive organization of the hepatic portal circulation (Mavrides et al., 2001), with an extrahepatic, unpaired trunk and two major intrahepatic branches giving origin to vessels of decreasing diameters: this hierarchical distribution serves as a basis for the functional segmentation of the liver, each liver segment being fed by a distinct major portal branch (Couinaud, 1999). It must be underlined that the definitive right and left intrahepatic portal veins have different origins: the right one is a remnant of the primitive right vitelline

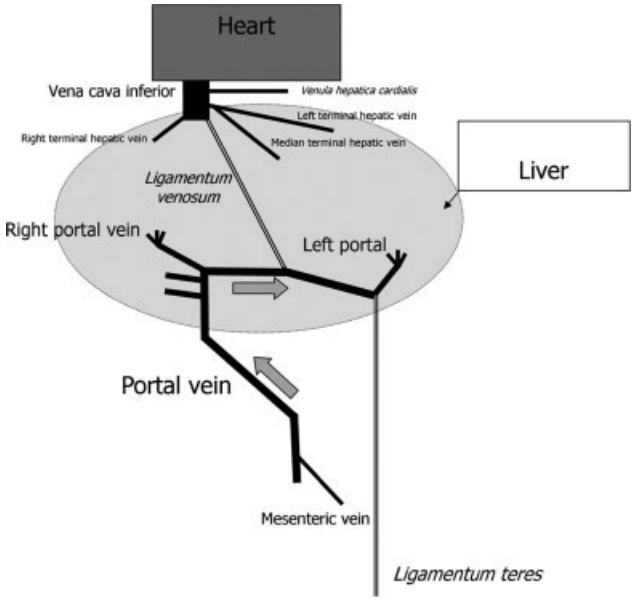


Fig. 6. The definitive vascular architecture of the liver, established at birth. The umbilical vein collapses and transforms into the *ligamentum teres*; the *ductus venosus* closes and is transformed into the *ligamentum venosum*. The portal vein becomes the unique afferent venous vessel of the liver; its right intrahepatic branch derives from the right vitelline vein whereas its left branch derives from the portal sinus and its tributaries. The arrows indicate the direction of the blood, which now flows freely into the portal venous system.

vein, whereas the left one derives from the portal sinus and its tributaries (Mavrides et al., 2001); this explains why the definitive left portal vein begins by an horizontal segment (corresponding to the portal sinus) followed by an oblique one (the so-called umbilical segment, corresponding to a part of the umbilical vein and to the *ramus angularis*; Gallego et al., 2002). The intrahepatic portal system will continue to grow and to branch out during childhood, in order to accommodate the progressive growth of the liver parenchyma.

Even if the umbilical circulation collapses at birth, the corresponding vessels do not completely disappear. Indeed, in the adult, the extrahepatic remnants of the umbilical vein and of the *ductus venosus* form well recognizable fibrous structures: the umbilical vein transforms into the *ligamentum teres hepatis*, running along the falciform ligament from the umbilical region to the transverse fissure separating the right and the left lobes of the adult liver, whereas the *ductus venosus* transforms into the *ligamentum venosum*, running in the continuity of the *ligamentum teres* until the inferior vena cava. Moreover, these structures retain some vascular patency: the repermeabilization of the umbilical vein is a well known consequence of portal hypertension.

EFFERENT VENOUS VESSELS

The development of the efferent venous vessels of the liver is at least as complex as that of its afferent venous vessels, but remains comparatively less well known. Only the process of formation of the terminal hepatic

veins has been reasonably well described (Dickson, 1957).

At the 4th gestational week, the subdiaphragmatic anastomosis connecting the more cranial portions of the two vitelline veins above the hepatic primordium (often called the subdiaphragmatic vestibulum) receives the blood from the sinusoidal plexus through two main vessels deriving from the corresponding segments of the right and left vitelline veins, after their disruption by the growing liver bud; these vessels are supplied by numerous vascular channels, or *venæ revehentes*, collecting the blood from the sinusoidal plexus. The anastomosis itself drains into the *sinus venosus* through the very last portion of the primitive vitelline veins. At this stage, all the efferent blood coming from the liver is, therefore, transported through two main vessels, which may be considered as the primitive terminal veins; they are known as the primary right and left hepatic veins (Dickson, 1957).

At the 5th gestational week, a profound remodeling of the efferent venous system of the liver takes place (Dickson, 1957). The right part of the subdiaphragmatic anastomosis enlarges and fuses with the cranial part of the primary hepatic vein to form the common hepatic vein. The corresponding *vena revehens*, deriving from the caudal segment of the primitive right vitelline vein, persists and will form the definitive right terminal hepatic vein, which drains part of the right lobe (Dickson, 1957; Mavrides et al., 2001). In contrast, the vascular segments deriving from the left vitelline vein, including the primary left hepatic vein, collapse and disappear. Their functional role is taken up by numerous vascular channels which drain part of the right lobe and most of the left lobe and open directly into the common hepatic vein. Progressively, these vascular channels will organize to form the secondary left terminal hepatic vein, which drains part of the left lobe (except for its cranial part, depending on an independent vessel, the *venula hepatica cranialis*), and the median terminal hepatic vein, which drains part of both lobes (Dickson, 1957; Mavrides et al., 2001). Usually, the middle and the left terminal hepatic veins have a common terminal segment which opens either into the common hepatic vein or into the *ductus venosus* before its termination. This is only a general scheme: many individual variations exist in the number and arrangement of the terminal hepatic veins. After this stage, little change will occur in the organization of the terminal hepatic veins. The main transformation in the efferent vessels of the liver occurs at the 8th gestational week, when the common hepatic vein merges into the developing vena cava inferior.

In contrast to the development of extrahepatic efferent vessels, that of the intrahepatic collecting veins feeding the terminal hepatic veins is very poorly known. In the adult, the first level of the efferent vascular system is the centrilobular vein, which drains all the blood circulating in the sinusoidal network of one hepatic lobule. Centrilobular veins drain into collecting veins of increasing diameters, which eventually form large vascular channels connected with one of the terminal veins. This complex organization seems to progressively emerge in parallel with the progressive establishment of the lobular architecture of the liver, which begins during the fetal period and will be achieved only after birth. In accordance with this concept, we have previously shown that primitive centrilobular veins could be identified at the 10th gestational week, whereas primitive collecting veins were not detectable before the 12th gestational week (Gouysse et al., 2002). In the rat, it has been shown that the efferent venous vessels of the fetal liver are lined by fenestrated endothelial cells, like those of sinusoids, and acquire their definitive, continuous endothelial lining only in the postnatal life (Barbera-Guillem et al., 1986); this suggests that the first levels of the efferent venous system may derive from previous sinusoid vessels. Comparable data are not available for the human liver; however, according to our immunohistochemical studies, the endothelial cells lining the central veins display a chronological sequence of phenotypic changes, which may suggest a process of progressive differentiation during liver development (Gouysse et al., 2002).

ARTERIAL SUPPLY OF THE LIVER

In the adult liver, the arterial supply of the liver mainly depends on the hepatic artery, which is a branch of the celiac trunk (numerous anatomical individual variations are, however, encountered). The extrahepatic segment of the hepatic artery is an unpaired vessel closely associated with the extrahepatic segment of the portal vein until the hilum. Within the liver parenchyma, the hepatic artery gives origin to branches of decreasing diameters which follow the distribution of portal vessels. The intraportal branches of the hepatic artery supply several capillary networks (Takasaki and Hano, 2001), especially the periportal capillary plexus, the peribiliary capillary plexus, the *vasa vasorum* associated with large intrahepatic vessels and the capillaries of the Glisson's capsule. Moreover, part of the arterial blood is also supplied to the sinusoidal network, even if the exact anatomical nature of the connections between the arterial network and sinusoids remains somewhat elusive and controversial (Oikawa et al., 1999; Oda et al., 2000, 2003; Pannarale et al., 2007).

Hepatic Artery: Development and Relations to Other Hepatic Vessels

The hepatic artery enters the game much later than the venous vessels, which preexist to the hepatic primordium. Indeed, it is not before the 8th gestational week that the first offshoot of the hepatic artery, coming from the celiac trunk, is visible in the hepatic hilum, near the extrahepatic fetal portal vein. At the 10th gestational week, the first intrahepatic arterial radicals are visible within the liver parenchyma, along the large intrahepatic branches of the portal system (Gouysse et al., 2002). At this stage, their distribution is restricted to the central regions of the fetal liver. They progressively extend, always following the intrahepatic branches of the portal system, until the periphery of the liver which they reach at approximately the 15th gestational week (Gouysse et al., 2002). They likely continue to grow until the liver has reached its definitive size.

The development of the arterial circulation of the liver is, therefore, closely dependent on that of the intrahepatic portal system, which behaves as a frame for the progressive development of the arterial supply of the liver. The development of the arterial circulation of the

liver is also closely coordinated with the development of the biliary system (Desmet, 1991). The first arterial radicals become detectable in the portal tracts when these spaces become surrounded by the ductal plate, a transient structure formed by the precursors of epithelial biliary cells and which will give origin to the definitive bile ducts. The timing of ductal plate formation is the same as that of the emergence of intrahepatic arterial radicals: approximately 10 weeks in the peri-hilar area and approximately 15 weeks in the peripheral areas of the liver. It is, therefore, tempting to hypothesize a link between the formation of the ductal plate and the growth of arterial radicals. This is supported by the demonstration that ductal plate cells are able to synthesize large amounts of VEGF, a potent pro-angiogenic factor (Gouysse et al., 2002). In the future, it will be interesting to determine whether ductal plate cells are able to secrete the guidance molecules, such as the members of the netrin and semaphorin families, which have been shown to play an important role in arteriogenesis, i.e., the migration and growth of arterial vessels (Lu et al., 2004).

Arterial-Fed Capillary Plexuses of the Portal Space: The Example of the Peribiliary Plexus

In the adult liver, the peribiliary plexus is a dense capillary network surrounding all the bile ducts and fed by a terminal branch of the adjacent hepatic artery (Takasaki and Hano, 2001). The timing of the development of the peribiliary plexus is well known, thanks to the works of Terada and Nakanuma (Terada et al., 1989; Terada and Nakanuma, 1993). The formation of the peribiliary plexus is closely associated with the emergence of biliary ducts from the ductal plate (Desmet, 1991). At the very early stage, no capillary is visible along the emergiong ductal plate: only isolated endothelial cells, interpreted as angioblasts, can be detected by immunohistochemical techniques (Terada and Nakanuma, 1993; Gouysse et al., 2002). Capillary vessels then appear along the maturing ductal plate and begin to multiply and concentrate around the emerging bile ducts resulting from the remodeling of the ductal plate. The peribiliary plexus is well identifiable at the 15th gestational week in the central areas of the liver and at the 25th in the peripheral areas. The growth of the peribiliary plexus continues during the whole intrauterine period and even after birth, since it has been shown that the definitive architecture of the peribiliary plexus is only achieved at approximately 15 years (Terada and Nakanuma, 1993). The details of the development of the other intraportal capillary networks, including the periportal plexus and the loosely arranged capillary vessels found all over the portal spaces, are poorly known but their formation likely follows the same general principles.

The most controversial issue raised by the development of the peribiliary plexus lies in its origin. Two hypotheses have been proposed (Terada and Nakanuma, 1993): (1) according to the first hypothesis, the peribiliary plexus derives from small branches coming from the arterial radicals which progressively extend into the portal spaces: this hypothesis is supported by the anatomical observations made in the adult liver and by the strong temporal relationship observed between the two processes; (2) according to a second hypothesis, the development of the peribiliary plexus is independent of the formation of the intrahepatic arterial supply: the capillary vessels which will eventually form the peribiliary plexus et connect with the arterial radicals are formed by vasculogenesis, i.e., not from preexistent vessels but from endothelial cell precursors present in the portal spaces, as suggested by immunohistochemical studies (Terada and Nakanuma, 1993; Gouysse et al., 2002).

SINUSOIDS

Hepatic sinusoids are, by far, the most differentiated vessels present in the adult liver. They display some unique structural characteristics and their endothelial lining cells present with a highly distinctive and very specific phenotype. Briefly, hepatic sinusoids run from the periphery of the hepatic lobule to the centrilobular vein, between the hepatocyte plates (Fig. 7). They ensure a unidirectional perfusion of the hepatic lobule, essential for the acquisition and maintenance of the functional heterogeneity of hepatocytes and of the metabolic zonation of the hepatic parenchyma. Sinusoids are fed by the portal system through direct anatomical connections: it is usually assumed that sinusoids originate from the terminal venules, which are small vessels coming from the most distal portal branches and running along the periphery of the hepatic lobules. The sinusoidal blood is also enriched in arterial blood, but, as said before, the nature, localization(s) and extent of the connections between hepatic sinusoids and the arterial circulation remain controversial. Some authors, using morphological and intravital microscopical techniques, have reported on the existence of direct anatomical connections between the terminal hepatic arterioles and the sinusoids, in the periportal area of the hepatic lobule (Oda et al., 2000, 2003) or along the whole periphery of the hepatic lobule (Takasaki and Hano, 2001). In contrast, other authors have been unable to detect and locate such direct anastomoses (Oikawa et al., 1999; Pannarale et al., 2007). However, important interspecies differences in the distribution of terminal hepatic arteries are known to exist (Del Rio Lozano and Andrews, 1966): observations made in mouse or rat livers may not be transferable to the human liver.

The sinusoidal wall controls the large amount of bidirectional exchanges occurring between hepatocytes and the blood. To fulfill this role, it presents distinctive structural and functional adaptations (Wisse et al., 1996). The endothelial lining of hepatic sinusoids is discontinuous and formed by fenestrated endothelial cells, containing numerous intracytoplasmic pores behaving as a sieve; adjacent endothelial cells present only weak intercellular junctions and are separated, at least after tissue fixation, by large extracellular fenestrations. The endothelial lining is separated from the adjacent hepatocytes by a space known as the perisinusoidal or Disse's space. No organized subendothelial basement membrane is visible along the endothelial lining. Nevertheless, the perisinusoidal space contains large amounts of extracellular matrix. The composition of the perisinusoidal matrix is highly specific: it is devoid of classic laminins, but rich in collagens (including collagens I, III, IV, VI, and XVIII), fibronectin, tenascin, and syndecans (Couvelard et al., 1996, 1997). In addition to endothelial cells, the sinusoidal wall contains several other resident cell popu-

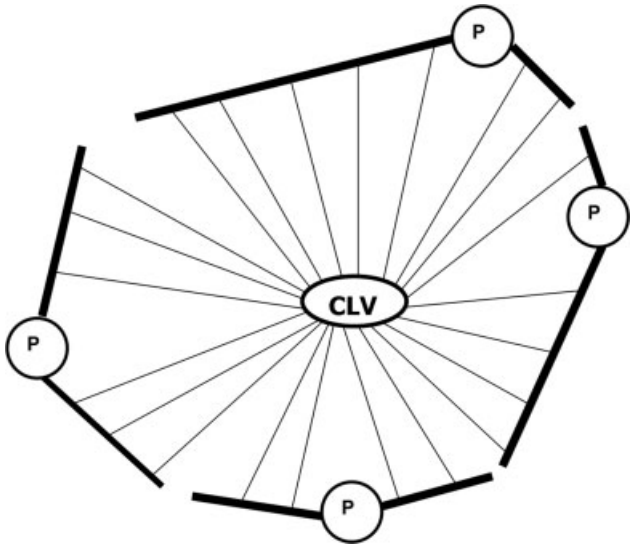


Fig. 7. Schematic representation of the general architecture of the sinusoidal plexus in the hepatic lobule. Sinusoids originate from the terminal venules, coming from the distal branches of the portal vein (PV) and located at the periphery of the lobule. They converge toward the centrilobular vein (CLV), in which their blood content is drained.

lations: (1) Kupffer cells, a population of differentiated intravascular macrophages, adherent to the luminal face of sinusoidal endothelial cells, (2) stellate or Ito cells, a population of mesenchymal, pericyte-like cells, located in the perisinusoidal space; they combine several distinctive functional properties, such as retinol storage and processing, extracellular matrix production and contractile capacities; they contribute to the regulation of the sinusoidal blood flow and to the synthesis of the perisinusoidal matrix (this role becomes essential during liver fibrogenesis); (3) pit cells, a population of resident lymphocytes.

In addition to their highly distinctive structural properties, sinusoidal endothelial cells are characterized by a very specific immunophenotype, which distinguishes them from all other endothelial cell populations in the body (Lalor et al., 2006). They lack several ubiquitous microvascular endothelial markers, such as CD34 (another typical endothelial cell marker, PECAM-1 or CD31 is expressed only at low levels, which may be undetectable in some conditions); in contrast, they express distinctive markers, such as CD4, ICAM-1, receptors for the Fc fragment of immunoglobulins, and the lipopolysaccharide-binding protein receptor; recent studies have pointed out to the specific expression of several hyaluronan receptors, such as LYVE-1 (also present in lymphatic endothelial cells; Mouta Carreira et al., 2001) and Stab-2 (which seems more specific for liver sinusoidal endothelial cells; Nonaka et al., 2007).

The structural and phenotypic properties of hepatic sinusoidal endothelial cells are usually interpreted as adaptations to their particular microenvironment and to their distinctive functions (Lalor et al., 2006). They are, therefore, expected to be progressively acquired during the embryonic and fetal stages of liver development. It must be, however, emphasized that the microenvironment and functions of sinusoids are quite different in the fetal liver and in the adult liver. The most striking example is the hematopoietic function of the fetal liver, which is the main hematopoietic organ of the body during most of the pregnancy. The production of hematopoeitic cells takes place along the hepatocyte plates and mature blood cells are released into the blood by migrating through the wall of the fetal sinusoids. The hematopoietic function of the liver declines only at the 7th month of gestation and entirely disappears during the perinatal period (it may be reactivated in postnatal life, in response to functional impairment of the bone marrow, which, in the meantime, has become the only hematopoietic organ of the body).

Several questions, therefore, arise about the development of hepatic sinusoids during liver organogenesis: (1) Which is their origin? (2) How do they adapt to their specific functions during the fetal life and when do they acquire their specific adaptations for adult life? (3) How and when do they acquire the highly specific characteristics observed in the adult liver?

Origin of Hepatic Sinusoids

Ontogenetically, hepatic sinusoids appear before all the other hepatic vessels. Their precursors can be recognized in between the growing cords of hepatoblasts, which begin to invade the mesenchyme of the *septum transversum* at the 4th week of gestation. The hepatic primordium is, therefore, formed not only by hepatic trabeculae but also by the intervening capillary vessels and their mesenchymal embedding. Two mutually non exclusive hypotheses have been proposed to explain the origin of hepatic sinusoids: (1) according to the first hypothesis, hepatic sinusoids derive from the pre-existing capillary vessels of the *septum transversum*, surrounded by the rapidly growing hepatocyte cords; (2) according to the second hypothesis, hepatic sinusoids result from the disruption of the pre-existing vitelline veins by the expanding hepatic primordium. According to both concepts, hepatic sinusoids are supposed to derive from pre-existing vessels. This suggests two important points: (1) the growth of sinusoids results from a process of angiogenesis, not vasculogenesis; (2) the acquisition of their specific structural and functional characteristics is not the result of a de novo process but involves the transdifferentiation of previous vessels, likely under the influence of signals coming from adjacent hepatocytes. It is unclear how early sinusoids grow and multiply to follow the rapid increase in size of the fetal liver; it has been suggested that, at least in avian embryos, the endothelial lining of the sinusoidal wall may incorporate cells of mesenchymal origin coming from precursors located in the *septum transversum* (Perez-Pomares et al., 2004); it has also been shown that the rodent fetal liver contains a population of angioblasts able to give origin to differentiated endothelial cells and to new vessels (Cherqui et al., 2006); recent works have shown that the human liver contains "hemangioblast" progenitors able to differentiate along either the hematopoietic or the endothelial lineages under the influence of hepatocytes (Bordoni et al., 2007).

Recent data indicate that the vascular component of the hepatic primordium has not a merely passive role in the early steps of liver organogenesis, as was thought before, but, in contrast, plays an essential role in the

induction of liver differentiation (Lammert et al., 2003). By using transgenic mice deficient in VEGFR2 (one of the receptors of the angiogenic factor VEGF), in which endothelial cell progenitors are unable to differentiate into blood vessels, it has been shown that liver development can progress until the formation of a multilayered epithelium on the ventral face of the foregut, but that the migration of hepatocytes in the *septum transversum* is impaired, resulting in the absence of a fully developed hepatic primordium (Matsumoto et al., 2001). In the same way, the growth of liver epithelium has been impaired in vitro, when vessel formation was inhibited in wild-type liver buds using angiogenesis inhibitors (Matsumoto et al., 2001). This set of experiments clearly demonstrates that endothelial cell signals, coming from developing vessels, are necessary for the initial steps of liver organogenesis. Candidates to the role of developmental inducers include extracellular proteins, such as members of the BMP (bone morphogenetic proteins) family, like BMP4 [in mice deficient for BMP4, liver epithelium differentiates but is unable to migrate into the *septum transversum* (Rossi et al., 2001)] and soluble growth factors, such as members of FGF [FGF8 has been shown to be required for liver development (Jung et al., 1999)] and TGFbeta families (Lammert et al., 2003).

First Stage of Sinusoidal Differentiation: From the 4th to the 12th Gestational Week

The progressive differentiation of sinusoids has been described in detail by using morphological techniques, including electron microscopy and, more recently, immunohistochemistry. At the 4th gestational week, the capillary vessels intervening between hepatocyte trabeculae present non specific characters: their endothelial lining is non fenestrated and continuous, the perisinusoidal matrix contains the components of a typical subendothelial basement membrane (including laminins), lining endothelial cells express the typical endothelial cell markers, such as CD34 and CD31, and lack the differentiated sinusoidal endothelial cell markers (Couvelard et al., 1996).

Between the 5th and the 12th gestational week, a first series of differentiation events takes place. The wall of sinusoids undergoes a dramatic structural remodelling. The perisinusoidal matrix looses the components of organized basement membranes, like laminins, between the 8th and 10th week of gestation (Couvelard et al., 1996). During the same period, the endothelial lining, initially continuous, becomes to acquire its typical cytoplasmic fenestrations (Enzan et al., 1997). The process of endothelial fenestration appears to be very progressive and the time point at which it is achieved is not clear from the literature data, which give figures ranging widely, from the 10th to the 17th gestational week (Makabe and Motta, 1980; Enzan et al., 1983; Machiarelli et al., 1988). In animals, the development of fenestrations in the endothelial lining of hepatic sinusoids is more precisely dated: 17 days of gestation in the rat and 15 in the mouse (Enzan et al., 1997); this roughly corresponds to 11 or 12 weeks of gestation in the human embryo. Interestingly, it has been shown in the rat that, at this stage, endothelial fenestrations are large, with a diameter greater than 250 nm (Barbera-Guillem et al., 1986); they are much larger than the typical endothelial fenestrae observed in the adult liver. Comparable data do not seem to be available for the human liver.

Recent immunohistochemical studies have demonstrated that, before the morphological remodeling of the sinusoidal wall described at the ultrastructural level, other differentiation events have taken place. Between the 5th and the 9th week of gestation, lining endothelial cells progressively loose the expression of the typical microvascular endothelial markers CD34 and CD31 (Couvelard et al., 1996; Fig. 8), which indicates a major remodeling of their intercellular junctional systems. Meanwhile, striking changes occur in the composition of the subendothelial matrix which shifts from that of a typical vascular matrix, initially rich in laminin and poor in tenascin, to that typical of the perisinusoidal matrix, poor in laminins and rich in tenascin (Couvelard et al., 1996, 1997); in the same time, the pattern of expression of cell–matrix adhesion receptors, such as integrins, undergoes marked changes in order to adapt to the composition of the novel perisinusoidal matrix (Couvelard et al., 1997).

It is interesting to note that the first stage of sinusoidal differentiation, including the development of endothelial fenestrations, the loss of the subendothelial basement membrane and the changes in the composition of the extracellular matrix, coincides with the beginning of the hematopoietic function of the liver, which starts at the 7th week of gestation and which will be maximal between the 3rd and the 7th months of gestation. It must be kept in mind that the structural characteristics acquired by the hepatic sinusoids during this period are strongly reminiscent of those observed in capillary vessels of other hematopoietic tissues, such as bone marrow sinusoids. It is, therefore, likely that the first stage of differentiation of hepatic sinusoids is necessary to adapt the intrahepatic vessels to the hematopoietic function of the fetal liver. This concept has an intriguing implication: that some of the characteristics of the postnatal sinusoids may be vestigial features of the fetal stage and not adaptations to adult hepatic functions.

The cellular and molecular mechanisms involved in the progressive differentiation of the sinusoidal endothelial lining of the fetal liver remain poorly understood. Microenvironmental factors, deriving from the perisinusoidal mesenchyme (Modis and Martinez-Hernandez, 1991; DeLeve et al., 2004) and/or from the differentiating hepatocytes (Yamane et al., 1994), are essential for the acquisition and maintenance of sinusoidal endothelial cell differentiation. Recent experimental studies suggest that members of the VEGF and TGFbeta families have an important regulatory role (Yamane et al., 1994; Yoshida et al., 2007). VEGF is especially important in the induction of the formation of the characteristic fenestrae of sinusoidal endothelial cells (Yokomori et al., 2003; Carpenter et al., 2005) and in their maintenance (Funyu et al., 2001).

Among the other sinusoidal cell populations, only the development and differentiation of stellate cells can be studied in detail (Bartok et al., 1983; Enzan et al., 1997): stellate cells, which likely derive from pre-existent mesenchymal cells of the *septum transversum* (Perez-Pomares et al., 2004), are first detectable between 8 and 10 weeks of gestation; they slowly differentiate, as shown the progressive increase in the number of intra-

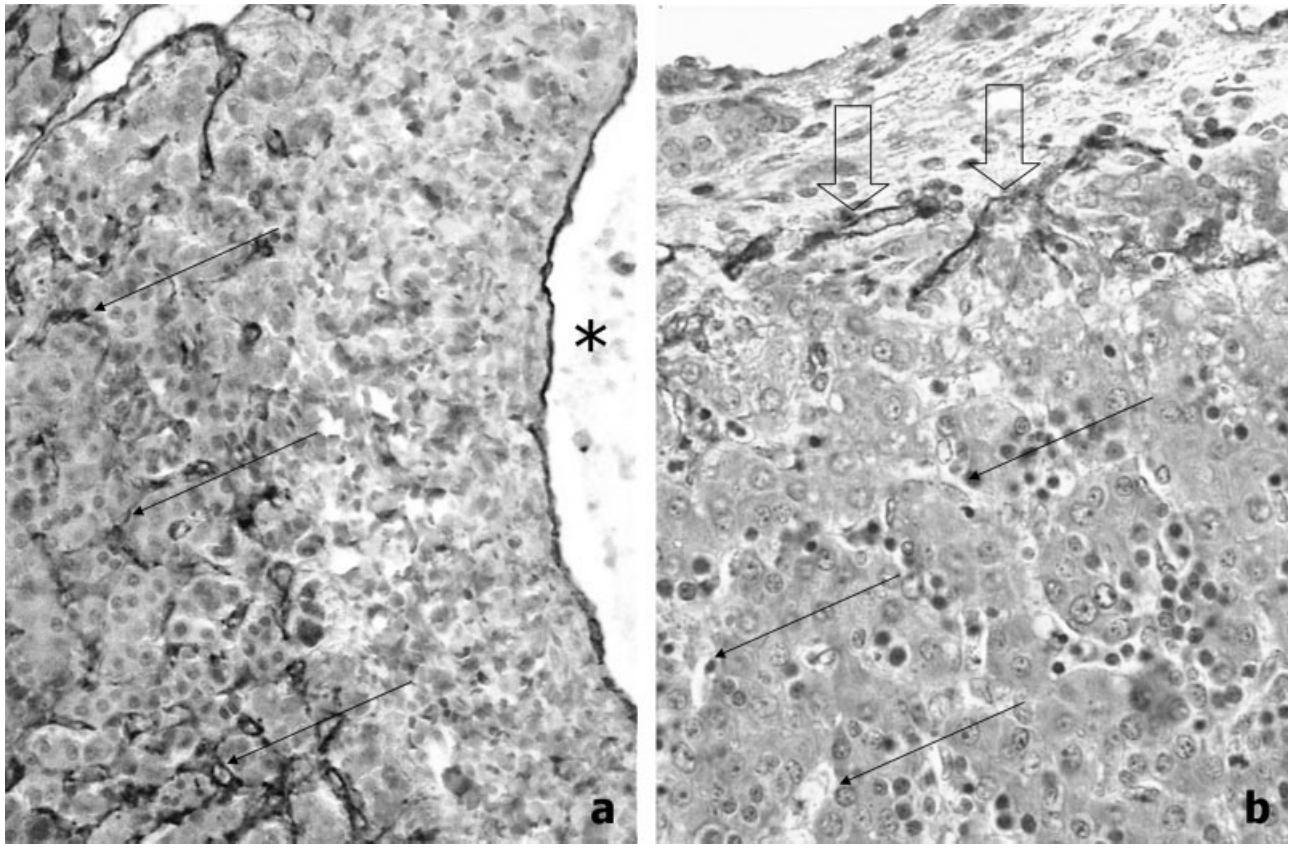


Fig. 8. Detection of the microvascular endothelial cell marker CD34 in the fetal liver at 5 weeks (a) and 12 weeks (b) of gestation. At 5 weeks, CD34 is detected along the endothelial lining of a large capillary vessel of the septum transversum (*) and along sinusoids (arrows). At 12 weeks, only small capillaries in a growing portal space (open arrows) are positive; hepatic sinusoids (arrows) are negative; note the active hematopoiesis present at this stage. Immunoperoxidase with nuclear counterstaining.

cytoplasmic lipid droplets, a characteristic feature of fully mature stellate cells. The development of Kupffer cells is more difficult to assess because of the physiological presence of hematopoietic cells until the 7th month of gestation (Naito et al., 1997).

Second Stage of Differentiation of Hepatic Sinusoids: From 12 to 20 Weeks of Gestation

Immunohistochemical studies have shown that, after the first phase of structural remodelling, a second phase of differentiation of the hepatic sinusoids takes place. It is characterized by the progressive acquisition of the specific sinusoidal endothelial markers, beginning with the appearance of detectable levels of CD4 at the 12th week of gestation and ending with the detection of the lipoplysaccharide-binding protein receptor (CD14) at the 20th week of gestation (Couvelard et al., 1996). The functions of all these markers are not precisely known, but most of them are involved in immune mechanisms, in the control of the homing and traffic of circulating leukocytes, and macromolecule clearance and uptake from the blood. This finding suggests that, in this second stage of differentiation, the endothelial lining of hepatic sinusoids progressively acquire all the characteristics observed during the postnatal life.

Third Stage of Differentiation of Hepatic Sinusoids: The Perinatal Period

In humans, there is very little information about the changes that may affect hepatic sinusoids in the late gestational stage, especially after the end of the hematopoietic activity, or at birth and during the immediate postnatal stage, after the major hemodynamic changes induced by the collapse of the umbilical circulation and the brutal increase in portal blood flow. In the rat, it has been shown that the definitive structural differentiation of the sinusoidal endothelial lining is achieved only in the perinatal period; there is a progressive disappearance of the large endothelial fenestrae, probably better adapted to the transmural migration of mature blood cells at the hematopoeitic stage, and a marked increase in the number of small fenestrae, better adapted for macromolecular transport (Barbera-Guillem et al., 1986). It will be interesting to perform comparable investigations in humans, in view of the major role played by the perinatal period in the establishment of the definitive architecture of the liver.

INTERACTIONS BETWEEN VASCULAR DEVELOPMENT AND LIVER GROWTH

It must be emphasized that all the processes of vascular development and differentiation which occur during the embryonic period and at later stages of gestation do not take place in a static, fixed environment, but in contrast, have to adapt within a highly dynamic and evolving background. The fetal liver is a rapidly growing organ. This finding is particularly striking at the very early stage of liver development. At 30 days of gestation, the hepatic primordium is formed only by the equivalent of two hepatic lobules; at the 35th day, in the 11-mm embryo, after the formation of the definitive fetal vascular architecture, 6 lobules only are present; very quickly, in the 20- to 24-mm embryos, no less than 700 portal branches can be counted, from the second to the sixth order (Mall, 1906 cited in Couinaud, 1999). The intrahepatic vessels have to adapt to this rapid increase in size and complexity: this suggests a rapid increase in their length and degree of branching. It is likely that the full development of intrahepatic vessels, venous and arterial, afferent and efferent, is not achieved until the end of liver growth, late in the postnatal life.

It must be also reminded that the growth of the human liver is allometric. If, in the early stages of liver development, the right and the left lobes of the liver are roughly equivalent, very soon, the right lobe becomes predominant whereas the left liver regresses. Moreover, the middle portion of the liver (corresponding to segment IV) will rapidly increase in size as compared with the other segments of the right liver, except for segment VI which, in humans as in the other primates, undergoes an important development. The allometric growth of the different liver segments modifies the respective lengths of the corresponding vessels and induces anatomical changes in the respective positions of the umbilical vein and of the portal vein (Couinaud, 1999).

Finally, the disposition of the large intrahepatic vessels has to adapt to the development of the lobular architecture of the liver, that is, the development of micro-anatomical units limited in periphery by several adjacent portal spaces and drained by one efferent, centrilobular vein. This suggests a strong interdigitation between the afferent portal vessels and the efferent terminal veins. In the early stages of liver development, the two systems do not interfere anatomically: afferent vessels retain a strict caudal localization and efferent vessels a strict cranial disposition until the 11-mm stage. The situation changes progressively until the 17-mm stage, in which portal vessels and terminal hepatic veins are found to interdigitate, in the same way as in the adult liver (Couinaud, 1999): this is a prerequisite for the functional organization of the fully developed hepatic lobule.

CONCLUSION

The vascular architecture of the adult liver is the product of a complex embryological history. The main lines of this history have been unravelled at the anatomical and histological levels. However, many questions remain to be answered, especially at the cellular and molecular levels, to better understand the complex interplay between the developing vessels and their environment which is responsible for this multistage process.

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